8.3 Area of 2D Figures

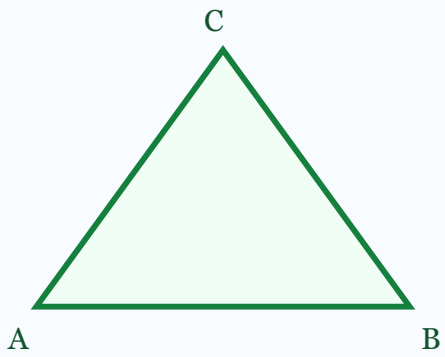
Every area formula for a polygon can be built from the rectangle. A triangle is exactly half of the parallelogram formed by duplicating it and rotating the copy $180°$, so $A = \frac{1}{2}bh$. A trapezoid can be doubled the same way into a parallelogram with base $b_1 + b_2$, giving $A = \frac{1}{2}(b_1 + b_2)h$. A rhombus or kite splits along its diagonals into four right triangles that reassemble into a rectangle with sides d_1 and d_2, cut in half: $A = \frac{1}{2}d_1 d_2$. A regular polygon splits into n congruent triangles from the center, each with height equal to the apothem, so its total area is $A = \frac{1}{2} \cdot \text{apothem} \cdot \text{perimeter}$. This worksheet also uses composite-figure decomposition and **Heron's formula**, $A = \sqrt{s(s-a)(s-b)(s-c)}$ with $s = \frac{a+b+c}{2}$, for triangles where only the three side lengths are known.

Examples

Example 1: Triangle area

A triangle has base 10 and height 6. Find its area.

Solution



Height is measured perpendicular to the chosen base.

Identify the base and height: $b = 10, h = 6$.

Apply the triangle area formula:

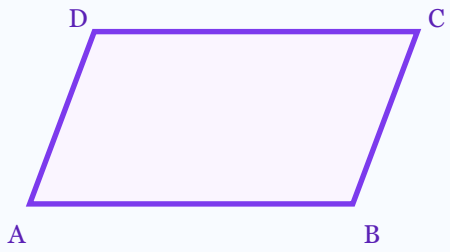
$$A = \frac{1}{2}bh = \frac{1}{2}(10)(6) = 30.$$

Answer: 30.

Example 2: Parallelogram area

A parallelogram has base 12 and height 5. Find its area.

Solution



Area = base × height, just like a rectangle sheared sideways.

Identify the base and height: $b = 12$, $h = 5$.

Apply the parallelogram area formula:

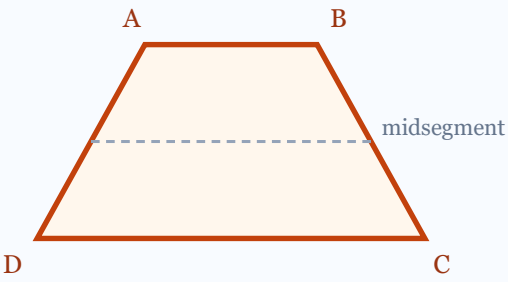
$A = bh = 12(5) = 60$.

Answer: 60.

Example 3: Trapezoid area (and why we average the bases)

A trapezoid has bases 8 and 14 and height 5. Find its area.

Solution



Two copies of the trapezoid form a parallelogram with base $b_1 + b_2$.

Two congruent copies of the trapezoid fit together to form a parallelogram with base $8 + 14 = 22$ and height 5, area $= 22(5) = 110$.

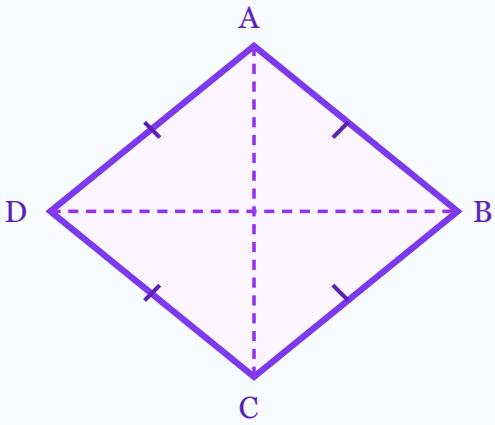
The trapezoid is half of that parallelogram: $A = \frac{1}{2}(8 + 14)(5) = \frac{1}{2}(22)(5) = 55$.

Answer: 55.

Example 4: Rhombus area from diagonals

A rhombus has diagonals 10 and 16. Find its area.

Solution



The diagonals split the rhombus into four right triangles that rearrange into a d_1-by-d_2 rectangle, cut in half.

Identify the diagonals: $d_1 = 10, d_2 = 16$.

Apply the rhombus area formula:

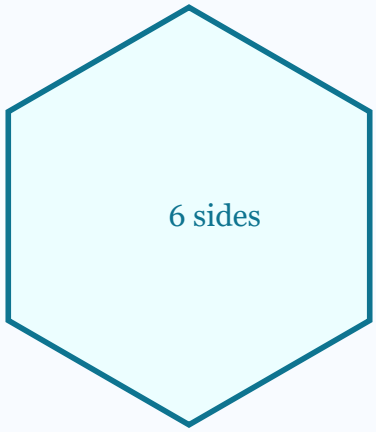
$$A = \frac{1}{2} d_1 d_2 = \frac{1}{2}(10)(16) = 80.$$

Answer: 80.

Example 5: Regular hexagon area (two ways)

A regular hexagon has side length 4. Find its area.

Solution



A regular hexagon splits into 6 congruent equilateral triangles.

Method 1 (six equilateral triangles): each has area $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(16) = 4\sqrt{3}$; total $= 6(4\sqrt{3}) = 24\sqrt{3}$.

Method 2 (formula): $A = \frac{3\sqrt{3}}{2}s^2 = \frac{3\sqrt{3}}{2}(16) = 24\sqrt{3}$.

Both agree.

Answer: $24\sqrt{3}$.

Example 6: Rectangle area

A rectangle has length 9 and width 4. Find its area.

Solution

Identify the length and width: $l = 9$, $w = 4$.

Apply the rectangle area formula:

$A = lw = 9(4) = 36$.

Answer: 36.

Example 7: Composite figure by decomposition

An L-shaped room is a 10×8 rectangle with a 3×4 rectangular corner removed. Find the room's area.

Solution

Area of the full rectangle: $10 \times 8 = 80$.

Area of the removed corner: $3 \times 4 = 12$.

Area of the L-shaped room: $80 - 12 = 68$.

Answer: 68 square units.

Example 8: Regular polygon from apothem and perimeter

A regular polygon has apothem 4 and perimeter 30. Find its area.

Solution

The polygon splits into n congruent triangles from the center, each with height = apothem and total base length = perimeter, so $A = \frac{1}{2} \cdot \text{apothem} \cdot \text{perimeter}$:

$A = \frac{1}{2}(4)(30) = 60$.

Answer: 60.

Example 9: Heron's formula — area from three sides

A triangle has sides 13, 14, and 15. Find its area using Heron's formula.

Solution

Semi-perimeter: $s = \frac{13 + 14 + 15}{2} = \frac{42}{2} = 21$.

$A = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{21(21-13)(21-14)(21-15)} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6}$.

$21 \cdot 8 = 168$; $168 \cdot 7 = 1176$; $1176 \cdot 6 = 7056$.

$A = \sqrt{7056} = 84$.

Answer: 84.

Example 10: Algebra with a kite's diagonals

A kite's diagonals are $3x$ and $4x$, and its area is 54. Find x and both diagonal lengths.

Solution

$A = \frac{1}{2}d_1 d_2 = \frac{1}{2}(3x)(4x) = 6x^2$.

$6x^2 = 54 \Rightarrow x^2 = 9 \Rightarrow x = 3$.

Diagonals: $3x = 9$ and $4x = 12$.

Check: $\frac{1}{2}(9)(12) = 54$. ✓

Answer: $x = 3$; diagonals 9 and 12.

Example 11: Application — mulching a garden bed

A trapezoidal garden bed has parallel sides 6 ft and 10 ft, and height 4 ft. Mulch costs \2.50$ per square foot. Find the area and the total cost to mulch the bed.

Solution

Area: $A = \frac{1}{2}(6 + 10)(4) = \frac{1}{2}(16)(4) = 32$ ft^2.

Cost: 32\times\2.50=$80. *< br >< b > Answer :< /b > area*=32*ft*^2; *cost*=80.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Triangle

Base 14, height 9. Area?

Answer: **Verification:**

Problem 2: Parallelogram

Base 11, height 6. Area?

Answer: **Verification:**

Problem 3: Trapezoid

Bases 7 and 13, height 4. Area?

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 4: Kite

Diagonals 9 and 12. Area?

Answer: **Verification:**

Problem 5: Rectangle

Sides 8 and 5. Area?

Answer: **Verification:**

Problem 6: Square

Side 12. Area?

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 7: Triangle

Base 20, height 7. Area?

Answer: **Verification:**

Problem 8: Rhombus

Diagonals 6 and 10. Area?

Answer: **Verification:**

Problem 9: Trapezoid

Bases 10 and 18, height 6. Area?

Answer: **Verification:**

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Problem 10: Regular polygon

Apothem 4, perimeter 30. Area?

Answer: **Verification:**

Problem 11: Parallelogram

Base 15, height 8. Area?

Answer: **Verification:**

Problem 12: Hexagon

Regular hexagon, side 2. Area?

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 13: Triangle

Base 9, height 12. Area?

Answer: **Verification:**

Problem 14: Challenge

Use Heron's formula to find the area of a triangle with sides 9, 10, 17.

Answer: **Verification:**

Problem 15: Challenge

An L-shaped patio is a 12×9 rectangle with a 5×3 rectangular corner removed. Find the remaining area.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. 63
2. 66
3. 40
4. 54
5. 40
6. 144
7. 70
8. 30
9. 84
10. 60
11. 120
12. $6\sqrt{3}$
13. 54
14. 36
15. 93 square units

Great work — keep practising!